

RM08

Shapes, Figure Counting, Mirror and Water Images, Embedded Figures

About 2-3 questions in every RAS Prelims paper. Counting triangles or squares comes almost every year (2023 and 2024 both carried one), mirror images came in 2023 and 2024, dice in 2024. The bank carries 45 questions on this chapter - do all of them, because speed here buys you time for the puzzle questions.

This is the picture chapter of the reasoning paper - count the triangles, name the mirror image, find the opposite face of the dice. There is nothing to memorise about the world here; there is a method for each type, and the method is short. Learn nine methods and the whole chapter becomes free marks.

Figures, Images and Counting

Counting triangles

- Count in layers: 1 region, then 2, then 3, then 4
- Rectangle + both diagonals = 8
- Rectangle + diagonals + one mid-line = 12
- Triangle + three medians = 16
- n cevians from one vertex = $C(n+2, 2)$

Counting squares & rectangles

- n x n grid squares = $n(n+1)(2n+1)/6$
- $3 \times 3 = 14$, $4 \times 4 = 30$, $8 \times 8 = 204$
- Rectangles = $C(m+1, 2) \times C(n+1, 2)$
- Chessboard rectangles = 1296

Mirror image

- Reflection about a VERTICAL axis
- Unchanged: A H I M O T U V W X Y
- Letter order always reverses
- Clock: true time = 11:60 minus image

Water image

- Reflection about a HORIZONTAL axis
- Unchanged: B C D E H I K O X
- Both mirror and water: H I O X, digits 0 and 8
- Clock: 18:30 minus the given time

Paper folding & punching

- n folds = 2^n layers
- One punch = 2^n holes
- k punches = $k \times 2^n$ holes
- Punch on a fold line gives fewer

Cube & dice

- Standard dice: opposite faces add to 7
- A face touches 4, is opposite the 5th
- Two views, same top: four sides in a cycle
- Painted cube: 8, $12(n-2)$, $6(n-2)^2$, $(n-2)^3$

Polygons & embedded figures

- Diagonals = $n(n-3)/2$
- Interior sum = $(n-2) \times 180$
- $n = 360/(180 - \text{interior angle})$
- Trapezium is the usual odd one out

What the paper actually asks - and where the marks leak

Everything in this chapter is a picture question. But the picture is always one of a small number of standard shapes, and RPSC describes it in plain words - 'a rectangle with both its diagonals drawn', 'two positions of the same dice'. So read the description, draw it yourself in the margin in five seconds, and then apply the method. Never try to hold the figure in your head.

- Counting - triangles, squares, rectangles inside a described figure.
- Mirror image - of a letter, a word, a number or a clock face.
- Water image - the same job, but the reflection is top-to-bottom.
- Paper folding and punching - how many holes after unfolding.
- Dice - which face is opposite which, from one or two positions.
- Painted cube - how many small cubes have 3, 2, 1 or 0 painted faces.
- Embedded or hidden figures, and odd-one-out among shapes.

Marks leak in two places only. One: the student counts the smallest triangles, finds four, and ticks 4 - forgetting that two small triangles joined together also make a triangle. Two: the student confuses the mirror set of letters with the water set. Both are cured by drill, not by cleverness. Fix the two and you will not lose a mark in this chapter again.

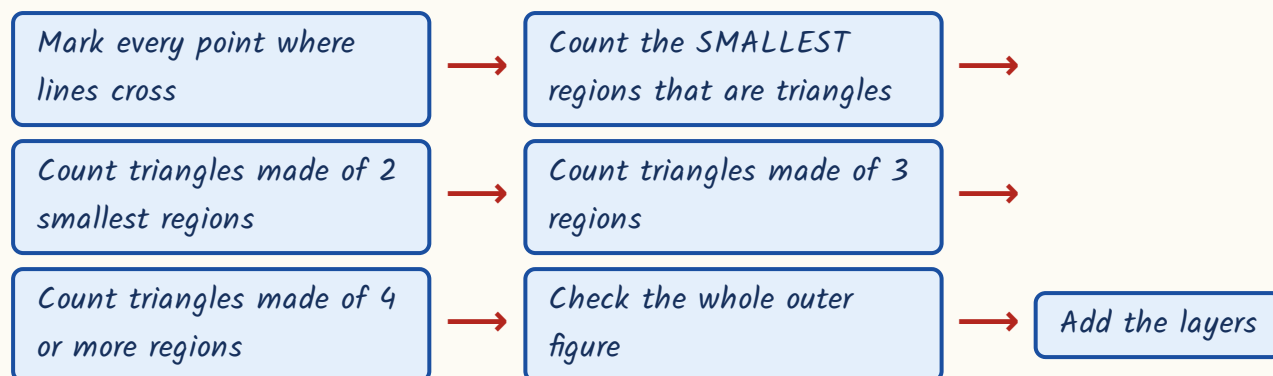
ASKED IN RAS

Asked in RAS Pre 2023 and 2024 - the rectangle with both diagonals (answer 8 triangles) and the 3 x 3 grid of squares (answer 14). RAS Pre 2024 also carried the rectangle with both diagonals plus one mid-line (12 triangles), the mirror-image letter question and a two-position dice question. This chapter repeats its own standard figures year after year.

Counting triangles - count in layers, never at random

There is exactly one reliable way to count triangles, and it is not to stare at the figure. You count in layers. First count the triangles made of a single smallest region. Then the triangles made of two of those regions joined. Then three. Then four. Then finally check the whole figure. Write the running total in the margin as you go - $4 + 4 = 8$, not a single number pulled out of the air.

The layered counting method - use it on every figure



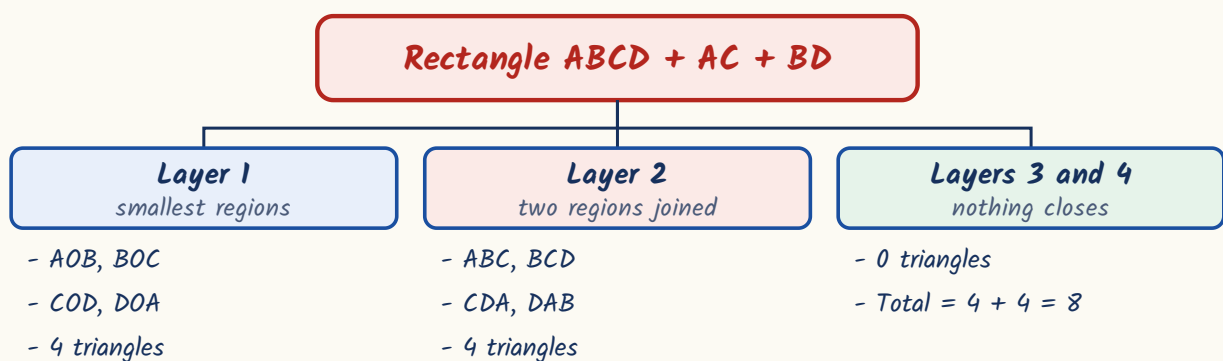
Standard figures - learn these numbers by heart

The figure described	Triangles	Squares
Rectangle or square + both diagonals	8	1 if it is a square
Rectangle + both diagonals + ONE mid-line	12	-
Square + both diagonals + BOTH mid-lines	16	5
Square + ONE diagonal + both mid-lines	6	5
Triangle + all three medians	16	-
Triangle with the three midpoints joined	5	-
Equilateral triangle, each side cut into 2	5	-
Equilateral triangle, each side cut into 3	13	-
Equilateral triangle, each side cut into 4	27	-

WORKED EXAMPLE 1 (RAS Pre 2023, 2024). A rectangle ABCD is drawn and both of its diagonals AC and BD are drawn. How many triangles are there in the figure in all?

- The two diagonals cross at the centre O - that is the only new point inside.
- Layer 1, smallest regions: $AOB, BOC, COD, DOA = 4$ triangles.
- Layer 2, two regions joined: $AOB + BOC$ gives ABC ; similarly $BCD, CDA, DAB = 4$.
- Layer 3, three regions joined: no triangle closes - 0.
- Layer 4, all four regions: that is the rectangle itself, not a triangle - 0.
- Total = $4 + 4 = 8$. Answer: 8.

Where the 8 triangles come from - rectangle with both diagonals



WORKED EXAMPLE 2 (RAS Pre 2024). A rectangle is divided by both its diagonals and also by the straight line joining the midpoints of its two longer sides. How many triangles are formed in all?

- The mid-line also passes through the centre O , so it cuts the top region and the bottom region into two each.
- Layer 1, smallest regions: 2 at the top + 2 at the bottom + 1 on the left + 1 on the right = 6.
- Layer 2, two regions joined: the top pair rejoins as AOB , the bottom pair as $COD = 2$.
- Layer 3: nothing closes - 0.
- Layer 4, half-rectangles: $ABC, ABD, BCD, ACD = 4$.
- Total = $6 + 2 + 4 = 12$. Answer: 12.

TRAP

Do not stop at the smallest regions. In Example 1 the smallest count is 4 and 4 is sitting there in the options waiting for you. The examiner puts the half-answer in the choices on purpose. Always ask: can two of these join into a bigger triangle? Can four?

Cevians, medians and the equilateral grid

When all the extra lines come out of ONE vertex of a triangle and land on the opposite side, you do not need to count at all. Every triangle in that figure is fixed by choosing two of the points on the opposite side and joining them to the apex. So count the points on that side and use the combination formula. With n lines drawn from the vertex, the opposite side carries $n + 2$ points and the answer is $C(n+2, 2)$.

Two shortcut families you can answer in five seconds

Configuration	Rule	Values
n lines from ONE vertex to the opposite side	$C(n+2, 2)$	$n=1: 3, n=2: 6, n=3: 10, n=4: 15$
Equilateral triangle, sides cut into n parts, lines parallel to all three sides	up-triangles + down-triangles	$n=2: 5, n=3: 13, n=4: 27$
Triangle with all three medians	fixed figure	16
Triangle with the three midpoints joined	4 small + 1 whole	5

WORKED EXAMPLE 3. In a triangle all three medians are drawn. How many triangles are there in the resulting figure?

- The three medians meet at the centroid G and cut the triangle into 6 smallest triangles of equal area.
- Layer 1: those 6.
- Layer 2, made of two small ones: $ABG, BCG, CAG = 3$.
- Layer 3, made of three small ones - each median splits the triangle in half: $ABD, ADC, BCE, BEA, CAF, CFB = 6$.
- Layer 6, the whole triangle = 1.
- Total = $6 + 3 + 6 + 1 = 16$. Answer: 16.

Counting squares and rectangles in a grid

A grid question is pure formula. For squares you add the squares of the numbers - a 3×3 grid holds $9 + 4 + 1$ squares because a 1-unit square can sit in 9 places, a 2-unit square in 4 places and a 3-unit square in 1. For rectangles you choose lines, not cells: any two of the vertical lines and any two of the horizontal lines bound exactly one rectangle.

Squares in a 3×3 grid, counted by size

Side 3 units

the whole square

1 place

Side 2 units

corner can sit in a 2×2 block

4 places

Side 1 unit

the small cells themselves

9 places

Grid formulas and the numbers worth memorising

Asked	Formula	Standard values
Squares of all sizes in an $n \times n$ grid	$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$	2×2 : 5, 3×3 : 14, 4×4 : 30, 5×5 : 55, 8×8 : 204

Asked	Formula	Standard values
Rectangles of all sizes in an $m \times n$ grid	$C(m+1, 2) \times C(n+1, 2)$	3x2 grid: $6 \times 3 = 18$
Rectangles on a chessboard	$C(9,2) \times C(9,2)$	$36 \times 36 = 1296$
Rectangles that are NOT squares	total rectangles minus squares	Chessboard: $1296 - 204 = 1092$

WORKED EXAMPLE 4. (a) RAS Pre 2023 - a square is divided into 9 equal smaller squares by two horizontal and two vertical lines. How many squares of all sizes are there? (b) How many rectangles of all sizes are there on a standard 8×8 chessboard, and how many on a 3×2 arrangement of small rectangles?

- (a) Squares of side 1 unit: $3 \times 3 = 9$.
- (a) Squares of side 2 units: the top-left corner has $2 \times 2 = 4$ possible positions, so 4.
- (a) Squares of side 3 units: the whole square = 1. Total = $9 + 4 + 1 = 14$.
- (a) Formula check: $n(n+1)(2n+1)/6 = 3 \times 4 \times 7 / 6 = 14$. Answer: 14.
- (b) A rectangle is fixed by 2 vertical lines and 2 horizontal lines.
- (b) A 3×2 arrangement has 4 vertical and 3 horizontal lines: $C(4,2) \times C(3,2) = 6 \times 3 = 18$.
- (b) A chessboard has 9 vertical and 9 horizontal lines: $C(9,2) \times C(9,2) = 36 \times 36 = 1296$.

YAAD RAKHO

Do not confuse the two chessboard numbers. 204 is SQUARES, 1296 is RECTANGLES. RPSC puts both in the options of the same question. Squares add squares; rectangles choose lines.

Mirror images – the vertical-axis rule

A mirror stands vertically beside the object, so the reflection swaps left and right and leaves top and bottom alone. This is called lateral inversion. Two consequences follow, and both get asked. First, a single letter survives unchanged only if it has a vertical axis of symmetry. Second, in a word the ORDER of the letters always reverses - REASONING becomes GNINOSAER - whether or not the individual letters are symmetric.

The two invariant sets – learn them as blocks, not one by one

Reflection	Unchanged capitals	Also note
Mirror (vertical axis)	A H I M O T U V W X Y (11 letters)	Digits unchanged: 0, 8
Water (horizontal axis)	B C D E H I K O X (9 letters)	Digits unchanged: 0, 8
Unchanged in BOTH	H I O X	Plus digits 0 and 8
Word rule (mirror)	Reverse the letter order first, then mirror each letter	A word reads the same only if it is a palindrome of symmetric letters, e.g. TOOT

WORKED EXAMPLE 5 (RAS Pre 2024). (a) A clock seen in a mirror shows the time as 3:40. What is the correct time? (b) A wall clock actually shows 4:50 - what will its mirror image show? (c) In the word MOUNTAIN, how many letters remain unchanged in their mirror image?

- (a) Rule: true time and mirror time always add up to 12:00, so subtract from 11:60.
- (a) 11:60 minus 3:40 - minutes 60 - 40 = 20, hours 11 - 3 = 8. Correct time = 8:20.
- (b) The rule works both ways: image of 4:50 = 11:60 - 4:50 = 7:10.
- (b) Check: 4:50 + 7:10 = 12:00. Correct.
- (c) Mirror-invariant set is A H I M O T U V W X Y.
- (c) In M-O-U-N-T-A-I-N the letters M, O, U, T, A, I qualify; the two N's do not. Answer: 6.

ASKED IN RAS

Asked in RAS Pre 2023 and 2024 - 'which capital letter looks exactly the same in a vertical mirror' (answer T, from the set A H I M O T U V W X Y), and RAS Pre 2024 - the mirror clock showing 3:40 (answer 8:20). Both are one-line answers if the set and the 11:60 rule are on your fingertips.

Water images - the horizontal-axis rule

A water image is what you would see reflected in still water below the object. The axis is horizontal, so top and bottom swap and left and right stay put. Therefore a letter survives only if it has a horizontal axis of symmetry - and that is a completely different set from the mirror set. In a word the letter order does NOT reverse in a water image; each letter simply flips upside down in place.

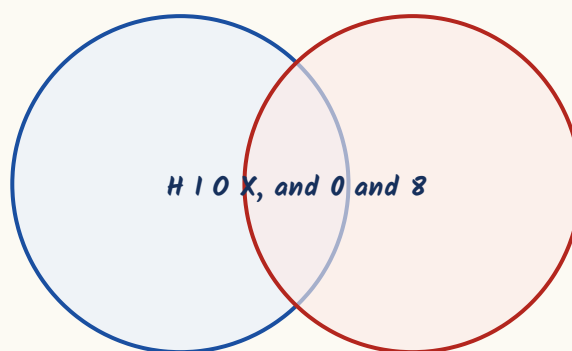
Mirror image vs water image - the single most confused pair in this chapter

Mirror image	Water image
Reflection about a VERTICAL axis	Reflection about a HORIZONTAL axis
Left and right swap; top and bottom stay	Top and bottom swap; left and right stay
Letter ORDER in a word reverses	Letter order stays the same
Survivors: A H I M O T U V W X Y	Survivors: B C D E H I K O X
Clock: 11:60 minus the given time	Clock: 18:30 minus the given time
Mirror stands beside the object	Water lies below the object

The two survivor sets and where they overlap

Mirror set - vertical axis

Water set - horizontal axis



Mirror only: A M T U V W Y. Water only: B C D E K. Confusing the two costs the mark.

WORKED EXAMPLE 6. (a) In the word CHOICE, how many letters remain unchanged in their water image? (b) Out of the capital letters A, B, H, M, O and X, how many remain unchanged in their water image?

- Water-invariant set: B, C, D, E, H, I, K, O, X.
- (a) Take C-H-O-I-C-E letter by letter: C yes, H yes, O yes, I yes, C yes, E yes.
- (a) All 6 letters survive, and since the order does not reverse the water image still reads CHOICE. Answer: 6.
- (b) From A, B, H, M, O, X: B yes, H yes, O yes, X yes = 4.
- (b) A and M are rejected - they are mirror-symmetric, not water-symmetric. Answer: 4.

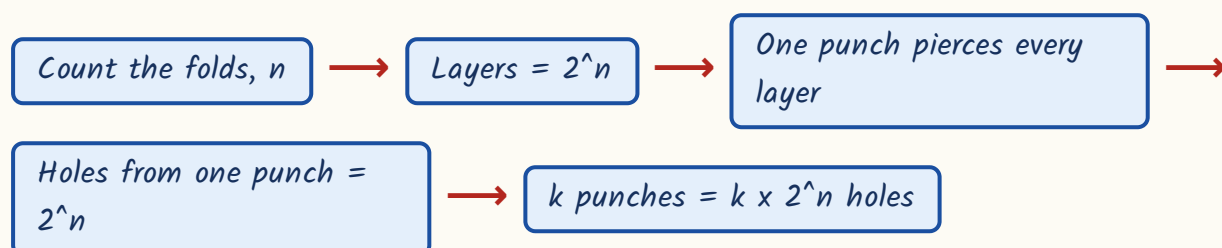
TRAP

Never use the 11:60 rule for a water clock. Mirror clock = 11:60 minus the time; water clock = 18:30 minus the time (subtract 12 hours if the answer crosses 12:00). And do not carry A and M into the water set - they are the classic wrong ticks.

Paper folding and punching

Fold a sheet in half and you have two layers. Fold again and you have four. Every fold doubles the layers, so after n folds there are 2 to the power n layers, and a single punch goes through all of them. The whole question type reduces to one line of arithmetic: holes = number of punches $\times$ 2 to the power n .

Folds to layers to holes



WORKED EXAMPLE 7. (a) A square sheet is folded exactly in half, then in half once more, and a single hole is punched at a point not on any fold. How many holes on unfolding? (b) The same sheet, two folds, but two separate holes punched. (c) Three folds, one punch.

- (a) Two folds, so layers = $2^2 = 4$. One punch pierces all 4 layers, giving 4 holes.
- (b) Layers still 4; each punch makes 4 holes, so $2 \times 4 = 8$ holes.
- (c) Three folds, layers = $2^3 = 8$, so one punch gives 8 holes.
- General: holes = $k \times 2^n$, where k is the number of punches and n the number of folds.

TRAP

The question says 'at a point not lying on any fold' for a reason. If the punch is made ON a fold line, the layers meet there and you get FEWER holes than 2^n - typically half. Read that clause before you multiply.

Dice and painted cubes

A cube has six faces. Any one face touches exactly four others and is opposite the fifth - that single sentence solves most dice questions. On a standard dice the opposite faces add up to 7, so the pairs are 1-6, 2-5 and 3-4. When the question gives you two positions of the same dice instead, you work out the cycle of the side faces.

A standard dice - the three opposite pairs

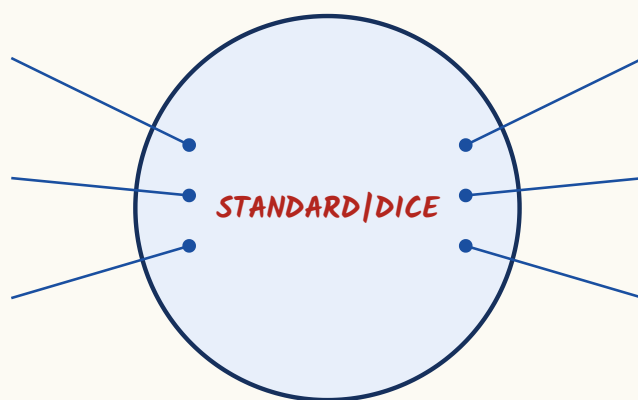
1 and 6

opposite pair, adds to 7

3 and 4

opposite pair, adds to 7

Two views, same top
the 4 sides form a cycle



2 and 5

opposite pair, adds to 7

Any one face

touches 4, opposite the 5th

Inside that cycle

opposites are two steps apart

Dice rules - which one applies to which question

What the question gives	Rule to apply	Result
'Standard dice' or 'ordinary dice'	Opposite faces sum to 7	Pairs 1-6, 2-5, 3-4
One face named as adjacent to four others	A face touches 4 and is opposite the 5th	The unnamed sixth number is the opposite
Two views with the SAME face on top	The other four lie in a cycle around it; opposites are two steps apart	Two pairs of opposites at once
Two views with ONE common face in different positions	Faces that never appear together in a single view are opposite	One pair of opposites

WORKED EXAMPLE 8 (RAS Pre 2024). Two positions of the same dice.

Position I: top face 1, front face 2, right face 3. Position II: top face 1, front face 4, right face 5. Which number lies opposite the face bearing 2?

- 1 is on top in BOTH views, so 2, 3, 4 and 5 are the four side faces and 6 must be at the bottom, opposite 1.
- Read each view clockwise from the front: view 1 gives the consecutive pair 2 then 3.
- View II gives the consecutive pair 4 then 5.
- The only cycle of the four side numbers containing 2-3 and 4-5 in that order is 2, 3, 4, 5 and back to 2.
- In a four-cycle the opposite faces are two steps apart: 2 opposite 4, and 3 opposite 5.
- Answer: 4 lies opposite 2.

Painted cube cut into n^3 smaller cubes - all six faces painted

Faces painted	Count	Where they sit
Exactly 3	8 always	The 8 corners, whatever n is
Exactly 2	$12(n - 2)$	Along the 12 edges, corners excluded
Exactly 1	$6(n - 2)^2$	The inside of each of the 6 faces
None	$(n - 2)^3$	The hidden inner block
At least 1	$n^3 - (n - 2)^3$	Everything except the inner block
Only two OPPOSITE faces painted	exactly one face: $2n^2$; two faces: 0	Top layer and bottom layer only

WORKED EXAMPLE 9. A cube painted on all six faces is cut into 64 identical smaller cubes. How many have exactly two faces painted, how many exactly one, and how many none at all?

- $64 = 4^3$, so $n = 4$ and $n - 2 = 2$.
- Three faces painted = 8 (the corners, always 8).
- Two faces painted = $12(n - 2) = 12 \times 2 = 24$.
- One face painted = $6(n - 2)^2 = 6 \times 4 = 24$.
- No face painted = $(n - 2)^3 = 2^3 = 8$.
- Check the identity: $8 + 24 + 24 + 8 = 64$. The four counts must always add back to n^3 .

YAAD RAKHO

Always run the addition check - $8 + 12(n-2) + 6(n-2)^2 + (n-2)^3$ must equal n^3 . It takes four seconds and catches every arithmetic slip. And remember the corner count is 8 for a 27-cube, a 64-cube or a 1000-cube alike; a cube has 8 corners, full stop.

Polygons, embedded figures and odd-one-out

The tail end of this chapter is three small types. Polygon questions are pure formula. Embedded-figure questions ask which of four complex figures hides a given simple shape - the method is to count the line segments and closed regions of the target shape and then look for exactly that many in each option, matching piece by piece rather than eyeballing. Odd-one-out among shapes usually turns on a definition, most often the parallel-sides definition of a quadrilateral.

Polygon formulas that get asked directly

Asked	Formula	Example
Number of diagonals	$n(n - 3)/2$	Octagon: $8 \times 5 / 2 = 20$
Sum of interior angles	$(n - 2) \times 180$ degrees	Hexagon: $4 \times 180 = 720$
Each exterior angle (regular)	$360/n$	Decagon: 36 degrees

Asked	Formula	Example
Sides from the interior angle	$n = 360/(180 - A)$	$A = 144: n = 360/36 = 10$
Each interior angle (regular)	$(n - 2) \times 180 / n$	Pentagon: 108 degrees

WORKED EXAMPLE 10. (a) Each interior angle of a regular polygon measures 144 degrees - how many sides? (b) How many diagonals does a regular octagon have? (c) Three of these are alike and one is different: rhombus, square, trapezium, rectangle. Which is the odd one out?

- (a) Exterior angle = $180 - 144 = 36$ degrees; number of sides = $360/36 = 10$.
- (a) Same thing in one step: $n = 360/(180 - A) = 360/36 = 10$ sides.
- (b) Diagonals = $n(n - 3)/2$ with $n = 8$, so $8 \times 5 / 2 = 20$ diagonals.
- (c) A rhombus, a square and a rectangle are all parallelograms - both pairs of opposite sides parallel.
- (c) A trapezium has only ONE pair of parallel sides, so the trapezium is the odd one out.

REVISION RECAP

1. Count triangles in layers - smallest regions, then 2 joined, then 3, then 4, then the whole figure.
2. Rectangle or square with both diagonals = 8 triangles; with one mid-line added = 12.
3. Square with both diagonals and both mid-lines = 16 triangles; with one diagonal and both mid-lines = 6.
4. Triangle with all three medians = 16 triangles; with the three midpoints joined = 5.
5. n lines from one vertex to the opposite side give $C(n+2, 2)$ triangles - $n=2$ gives 6, $n=3$ gives 10.
6. Equilateral triangle with sides cut into n parts: 5 triangles for $n=2$, 13 for $n=3$, 27 for $n=4$.
7. Squares in an $n \times n$ grid = $n(n+1)(2n+1)/6$ - remember 14, 30 and 204.
8. Rectangles in an $m \times n$ grid = $C(m+1, 2) \times C(n+1, 2)$; chessboard = $36 \times 36 = 1296$, of which 204 are squares.

9. Mirror = vertical axis; the survivors are A H I M O T U V W X Y and the letter order reverses.
10. Water = horizontal axis; the survivors are B C D E H I K O X and the order does not reverse.
11. H, I, O, X and the digits 0 and 8 survive both reflections.
12. Mirror clock: true time = 11:60 minus the image; the two times always add to 12:00.
13. Water clock: 18:30 minus the given time, dropping 12 hours if the answer overshoots.
14. Paper folding: n folds give 2^n layers, so k punches give $k \times 2^n$ holes - fewer if the punch sits on a fold.
15. A cube face touches four faces and is opposite the fifth; on a standard dice opposites add to 7.
16. Two dice views with the same top: the four side numbers form a cycle and opposites are two steps apart.
17. Painted cube: 3 faces = 8, 2 faces = $12(n-2)$, 1 face = $6(n-2)^2$, none = $(n-2)^3$, and the four add to n^3 .
18. Only two opposite faces painted: exactly one face painted = $2n^2$, and no cube gets two painted faces.
19. Polygons: diagonals $n(n-3)/2$, interior sum $(n-2) \times 180$, sides from interior angle $A = 360/(180 - A)$.
20. In shape classification the trapezium is normally the odd one out - only one pair of parallel sides.